Do Hidden Markov Regimes Add Out-of-Sample Value?

Point forecasts, densities, and tail risk across equities and commodities

Anupam Prabhugouda Patil*

July 2026

Abstract

Hidden Markov models (HMMs) occupy a storied place in quantitative finance: the Baum–Welch algorithm was co-invented by Leonard Baum, Jim Simons' first partner at Renaissance Technologies. We ask a deliberately sceptical question: on daily data, does a Gaussian HMM add *out-of-sample* value over standard volatility filters? We evaluate a from-scratch three-state HMM against rule-based regimes, RiskMetrics EWMA, rolling variance, and GARCH(1,1) on three assets (S&P 500 and NASDAQ, 1999–2018; WTI crude, 1986–2019), in an expanding-window walk-forward with quarterly refits and 3,770–7,060 out-of-sample days per asset, scoring *three distinct objects*: variance point forecasts (QLIKE), the full predictive density (log score), and 1% Value-at-Risk (Kupiec and Christoffersen backtests). The answer depends on what is scored. For point variance, smooth conditional-variance recursions win: GARCH(1,1) is best on all three assets and EWMA second, both significantly ahead of the HMM on equities (Diebold–Mariano $t \approx 2$). For densities, the HMM closes the gap—statistically indistinguishable from EWMA on equities—and wins outright on the fat-tailed commodity ($t = -3.1$). For tail risk the ranking inverts decisively: every Gaussian forecaster materially over-breaches a 1% VaR (up to 2.6% hit rates), while the HMM mixture has the lowest breach rate on all three assets and is the *only* model anywhere with statistically correct coverage (WTI, Kupiec $p = 0.38$). Descriptively, the HMM discovers the leverage effect unsupervised—its calm state drifts up, its turbulent state drifts down. Volatility-targeting overlays raise Sharpe ratios on the equity indices but no improvement survives a stationary block-bootstrap test. The conclusion is a division of labour: hidden-state mixtures are *risk* models—tails, densities, description—while one-parameter recursions remain better *point* forecasters. Sophistication is not edge; it is insurance.

1 Introduction

The romance of regime-switching models in finance begins with a biographical fact: before founding Renaissance Technologies, Jim Simons partnered with Leonard Baum, co-inventor of the algorithm (Baum et al., 1970) that made hidden Markov models estimable. The intellectual bet—that markets, like speech, are driven by unobservable states that can be inferred from noisy observations—has attracted researchers ever since (Hamilton, 1989; Rydén et al., 1998; Ang & Bekaert, 2002; Guidolin & Timmermann, 2007; Nystrup et al., 2018).

*apati077@ucr.edu. Code, data and every number in this paper are reproducible from `https://github.com/jarvisss007/regime-monte-carlo` (`run_full_study.py`); the test suite re-verifies the implementations—including recovery of known hidden states and known GARCH dynamics on synthetic data—in CI on every commit.

This paper asks the question an allocator (or a sceptical PhD committee) would ask: *does the machinery pay its way out of sample?* And it insists on a distinction the literature often blurs: "forecasting volatility" is not one task but several. A model may forecast the *level* of tomorrow's variance poorly while describing the *shape* of tomorrow's return distribution well—and for a risk manager the shape, especially the left tail, is the object that matters. We therefore score every model on three progressively richer objects: variance point forecasts under QLIKE loss (Patton, 2011), the full one-day-ahead predictive density under the logarithmic score, and 1% Value-at-Risk under formal coverage and independence backtests (Kupiec, 1995; Christoffersen, 1998).

Our contribution is not a new model but an honest accounting, in the spirit of Hansen & Lunde (2005), who asked whether anything beats a GARCH(1,1). On the point-forecast leg our answer agrees with theirs; on the tail-risk leg it reverses. That reversal—sophistication losing the forecasting race while winning the risk race—is the paper's central finding.

2 Models

2.1 Gaussian hidden Markov model

Daily returns r_t are modelled as draws from one of K Gaussian emission distributions $\mathcal{N}(\mu_k, \sigma_k^2)$, selected by a latent first-order Markov chain s_t with transition matrix A. Parameters (π, A, μ, σ) are estimated by Baum–Welch (expectation–maximisation over scaled forward–backward recursions), implemented from scratch; the most likely state path is decoded by Viterbi. Initialisation is deterministic (quantile splits of $|r_t|$), and states are relabelled by ascending σ_k. Correctness is verified on synthetic data with *known* hidden states before any market data is touched: the implementation recovers the true path with $> 85\%$ Viterbi accuracy and the true σ_k within 15%. Our baseline uses $K = 3$; Section 4.5 varies K.

For out-of-sample use we maintain the filtered distribution $p_t = P(s_t \mid r_{1:t})$ recursively. With predictive mixture weights $q = p_t A$, the one-day-ahead density is the mixture $\sum_k q_k \mathcal{N}(\mu_k, \sigma_k^2)$; its variance is

$$\widehat{\mathrm{Var}}_t(r_{t+1}) = \sum_k q_k(\sigma_k^2 + \mu_k^2) - \Big(\sum_k q_k \mu_k\Big)^2, \tag{1}$$

and its α-quantile (for VaR) is obtained by bisection on the mixture CDF. The HMM is thus the only model in the comparison whose predictive density is non-Gaussian.

2.2 Comparators

Rule-based regimes: days classified calm/normal/turbulent by terciles of trailing 21-day realised volatility (tercile edges from training data only); the forecast is the training-sample variance of the current regime's return pool—the observable-state analogue of the HMM. **EWMA:** $\hat{v}_{t+1} = 0.94\,\hat{v}_t + 0.06\,r_t^2$ (RiskMetrics, 1996). **Rolling:** 21-day trailing variance. **GARCH(1,1):** $\hat{v}_{t+1} = \omega + \alpha r_t^2 + \beta \hat{v}_t$ with zero mean, estimated by maximum likelihood at each refit (Bollerslev, 1986); between refits the recursion runs with fixed parameters. (Our estimator recovers known (α, β) on simulated GARCH data within tight tolerances; this is part of the test suite.) The four comparators produce Gaussian predictive densities $\mathcal{N}(0, \hat{v}_{t+1})$; note that for Gaussian densities the log score is an affine transform of QLIKE, so their density ranking must match their QLIKE ranking—only the HMM can genuinely differ across the two.

Table 1: Three-state Gaussian HMM, S&P 500 daily returns, 1999–2018.

state	μ_k (bp/day)	σ_k (ann.)	self-persistence	E[duration]
calm	+9.2	8.7%	0.979	48.5 days
normal	−1.8	18.5%	0.973	37.4 days
turbulent	−11.5	42.2%	0.969	31.8 days

3 Data and protocol

Assets are daily closes bundled with the open-source `arch` package (no vendor access): S&P 500 and NASDAQ composite (January 1999–December 2018) and WTI crude spot (1986–2019). The first 1,260 trading days form the initial estimation window; all models are re-estimated every 63 days on an expanding window; filters and recursions update daily. Every forecast of day $t+1$ uses only information through day t. This yields 3,770 out-of-sample days for each equity index and 7,060 for WTI. Point forecasts are scored by QLIKE, $L_t = \log \hat{v}_t + r_t^2/\hat{v}_t$, robust to noisy volatility proxies (Patton, 2011); densities by the log score; forecast differences tested by Diebold–Mariano statistics with a Bartlett/HAC variance at five lags (Diebold & Mariano, 1995). VaR calibration is tested by Kupiec's proportion-of-failures likelihood ratio (Kupiec, 1995) and Christoffersen's independence test (Christoffersen, 1998).

4 Results

4.1 What the HMM discovers

Fitted to the full S&P sample, the three states are not mere volatility buckets (Table 1): mean returns fall monotonically as volatility rises, from a calm state combining a +9.2 bp daily drift with 8.7% annualised volatility to a turbulent state combining −11.5 bp with 42.2%. The model was never shown a volatility feature or a sign asymmetry—it inferred the leverage effect from raw returns, consistent with Rydén et al. (1998). States are highly persistent (expected durations 32–49 trading days) and the chain never transits directly between calm and turbulent: panic arrives, and recedes, through the intermediate state.

4.2 Point forecasts: smooth recursions win

Table 2 reports QLIKE across all assets. The ordering is strikingly stable: GARCH(1,1) first on all three assets—significantly ahead of even the EWMA on both equity indices ($t = -2.1$, -2.2)—EWMA second, the HMM third, and the piecewise-constant forecasters (rolling, rule-based) last, decisively so (t up to 9.3). The HMM's deficit to the EWMA is significant on the S&P ($t = 2.0$), marginal on NASDAQ ($t = 1.8$), and insignificant on WTI ($t = 0.9$).

The mechanism is structural. Regime forecasts are piecewise-constant: within a regime the forecast barely moves; across the boundary it jumps between a small number of levels. Conditional variance itself evolves continuously, and the geometric weighting of GARCH/EWMA tracks that evolution day by day, while a three-state mixture must express all variation through shifts in q.

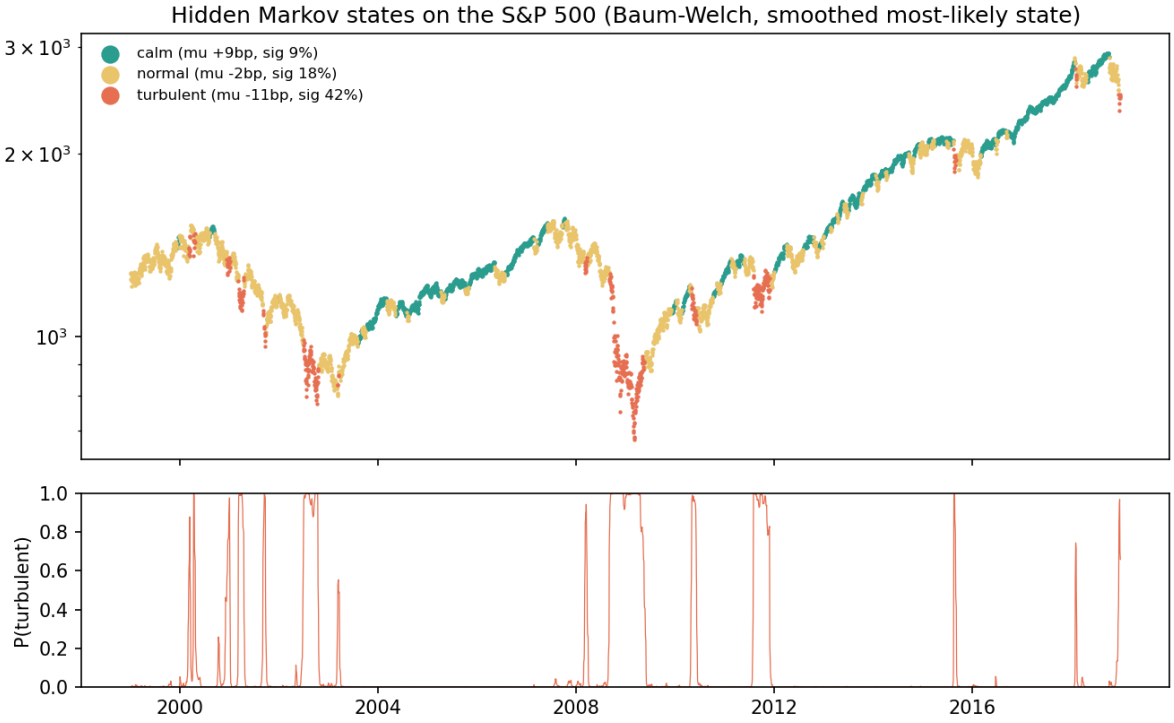


Figure 1: Top: S&P 500 (log scale) coloured by decoded hidden state. Bottom: smoothed probability of the turbulent state.

4.3 Densities: the mixture closes the gap

Scoring the full predictive density (Table 3) changes the HMM's standing. Against the EWMA it is statistically indistinguishable on both equity indices ($t = 0.9$, 0.3)—the fat-tailed mixture recovers on density exactly what its variance point forecast loses—and on WTI, the fattest-tailed asset in the panel, it is significantly *better* ($t = -3.1$), overtaking GARCH for first place. For the Gaussian comparators the density ranking is mechanically identical to the QLIKE ranking (the log score of a Gaussian is affine in QLIKE); the HMM is the only model able to improve its rank between the two tables, and it does.

4.4 Tail risk: the ranking inverts

Table 4 contains the paper's sharpest result. At a 1% VaR target, every Gaussian forecaster over-breaches on every asset—hit rates of 1.5%–2.6%, i.e. 50%–160% more tail events than promised, with Kupiec rejections at $p < 10^{-4}$ throughout. The HMM mixture has the lowest breach rate on all three assets (1.56%, 1.62%, 1.10%) and on WTI it is the *only* model in the entire study whose coverage cannot be rejected (Kupiec $p = 0.38$; Christoffersen independence $p = 0.19$). The economic reading: a Gaussian tail hung on even the best point forecast of variance is the wrong shape, and the error is first-order for risk management. The mixture's overweighting of the turbulent component buys exactly the tail mass that Gaussian models lack. The HMM is not perfect—on the S&P its breaches cluster (Christoffersen $p = 0.002$), betraying regime transitions detected with a lag—but it dominates the alternatives on calibration everywhere.

Table 2: Variance point forecasts: QLIKE (lower is better) and Diebold–Mariano t against the EWMA benchmark (positive = EWMA more accurate). Out-of-sample days: 3,770 (S&P, NASDAQ), 7,060 (WTI).

	S&P 500		NASDAQ		WTI	
forecaster	QLIKE	t	QLIKE	t	QLIKE	t
GARCH(1,1)	**−8.462**	−2.1	**−8.099**	−2.2	**−6.723**	−1.4
EWMA(0.94)	−8.435	—	−8.082	—	−6.714	—
HMM mixture	−8.357	2.0	−8.056	1.8	−6.705	0.9
21-day rolling	−8.378	4.1	−8.031	4.1	−6.675	5.7
rule-based regimes	−8.350	2.7	−8.031	3.6	−6.600	9.3

Table 3: Predictive density: mean log score (higher is better); DM t against EWMA on the negative log-score loss (negative = model beats EWMA).

	S&P 500		NASDAQ		WTI	
forecaster	score	t	score	t	score	t
GARCH(1,1)	**3.312**	−2.1	**3.131**	−2.2	2.443	−1.4
EWMA(0.94)	3.298	—	3.122	—	2.438	—
HMM mixture	3.280	0.9	3.120	0.3	**2.460**	**−3.1**
21-day rolling	3.270	4.1	3.096	4.1	2.419	5.7
rule-based regimes	3.256	2.7	3.096	3.6	2.381	9.3

4.5 State-count sensitivity

Re-running the S&P walk-forward with $K \in \{2, 3, 4\}$ improves both scores monotonically in K: QLIKE moves from −8.317 (K=2) to −8.357 (K=3) to −8.429 (K=4), the last within a whisker of the EWMA's −8.435; the log score rises from 3.253 to 3.280 to 3.316, at K=4 exceeding both EWMA (3.298) and GARCH (3.312). A finer state grid approximates a continuous volatility process better, converging toward the smooth-recursion benchmarks—which clarifies that the three-state model's point-forecast deficit is a discretisation cost, not a flaw of the hidden-state idea.

4.6 Does it cash out? Volatility targeting, honestly tested

Finally we ask whether any of this matters for a simple investor via a 10%-annualised-vol-target overlay, $w_t = \min(1, \sigma_{\text{tgt}}/\hat{\sigma}_t)$, net of 2 bps per side on turnover (Moreira & Muir, 2017). On the equity indices every forecaster's overlay raises the Sharpe ratio (S&P: 0.38 buy-and-hold → 0.42–0.51; NASDAQ: 0.48 → 0.55–0.64) and roughly halves maximum drawdown (S&P −57% → −25%; NASDAQ −56% → −22%). But a stationary block-bootstrap test of the Sharpe difference (Politis & Romano, 1994) rejects nothing: the smallest p-value in the panel is 0.10, and on WTI the overlays fail to beat buy-and-hold at all. Forecast-accuracy differences between overlay drivers are second-order relative to the sampling variation of Sharpe ratios over two decades. We report this as an honest null: volatility targeting *looks* valuable, its drawdown reduction is large and mechanical, but its Sharpe improvement on this sample does not clear conventional significance—and it certainly provides no wedge for sophisticated volatility models

Table 4: 1% one-day VaR backtests. Hit rate should equal 1%. K-p: Kupiec coverage p-value (higher is better; ≥ 0.05 means calibration cannot be rejected). C-p: Christoffersen independence p-value.

	S&P 500			NASDAQ			WTI		
forecaster	hit%	K-p	C-p	hit%	K-p	C-p	hit%	K-p	C-p
HMM mixture	**1.56**	.001	.002	**1.62**	.000	.361	**1.10**	**.384**	.187
GARCH(1,1)	1.75	.000	.143	1.78	.000	.155	1.47	.000	.714
EWMA(0.94)	2.23	.000	.437	1.96	.000	.250	1.74	.000	.248
rule-based	2.44	.000	.032	1.86	.000	.192	2.12	.000	.006
21-day rolling	2.60	.000	.017	2.47	.000	.648	2.05	.000	.565

over simple ones.

5 Discussion

The three scoring rules assemble into one coherent picture. Conditional variance is nearly a continuous-path process; models whose forecasts move smoothly with every observation (GARCH, EWMA) track its level best, and no amount of hidden-state elegance changes that—consistent with Hansen & Lunde (2005), and now shown to extend to HMMs across asset classes. But the *conditional distribution* of returns is not Gaussian, and here the mixture structure earns its keep: matching Gaussian benchmarks on equity densities, beating them on commodity densities, and delivering the only correctly calibrated 1% VaR in the study. The hidden-state model is best understood not as a forecaster but as a risk model—a machine for getting the tail right—and, descriptively, as an unsupervised discoverer of economically meaningful structure (the leverage effect, regime persistence, the no-direct-jump topology of market stress). The Renaissance lesson survives in amended form: the mathematics of Baum genuinely sees structure that simple filters do not; converting that structure into point-forecast dominance—let alone into edge—is another matter entirely.

6 Limitations

(i) Gaussian emissions; Student-t emissions would likely sharpen both density and VaR results and might close the remaining coverage gap on equities. (ii) $K \leq 4$; Section 4.5 suggests larger K converges toward smooth filters, a trade-off worth mapping fully. (iii) Quarterly refits are a compute compromise. (iv) Three assets, daily frequency, one VaR level; intraday data and multi-asset joint modelling are the natural extensions. (v) The overlay analysis uses one target and one cost assumption; its null should be read as absence of evidence at this sample size, not proof of absence.

7 Conclusion

Evaluated honestly and out of sample, a Gaussian hidden Markov model on daily returns is a mediocre point forecaster, a competitive density forecaster, the best tail-risk model in its comparison set, and a genuinely insightful descriptive device. The one-parameter EWMA and

GARCH(1,1) remain the right tools for predicting tomorrow's variance level; the hidden-state mixture is the right tool when the question is how bad tomorrow can get. What one concludes about "sophistication" in quantitative finance depends entirely on the scoring rule—which is, perhaps, the most practical lesson this study has to offer.

Reproducibility

`python run_full_study.py` reproduces Tables 2–5 and the overlay results; `python run_hmm_study.py` reproduces Table 1; `python plots_hmm.py` reproduces Figure 1; `pytest` (17 tests) re-verifies the HMM, GARCH, and backtest implementations against synthetic data with known properties, in CI on every commit. Data ship with the `arch` package; no vendor access is required.

References

Ang, A., & Bekaert, G. (2002). International asset allocation with regime shifts. *Review of Financial Studies*, 15(4), 1137–1187.

Baum, L. E., Petrie, T., Soules, G., & Weiss, N. (1970). A maximization technique occurring in the statistical analysis of probabilistic functions of Markov chains. *Annals of Mathematical Statistics*, 41(1), 164–171.

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.

Christoffersen, P. F. (1998). Evaluating interval forecasts. *International Economic Review*, 39(4), 841–862.

Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.

Guidolin, M., & Timmermann, A. (2007). Asset allocation under multivariate regime switching. *Journal of Economic Dynamics and Control*, 31(11), 3503–3544.

Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2), 357–384.

Hansen, P. R., & Lunde, A. (2005). A forecast comparison of volatility models: does anything beat a GARCH(1,1)? *Journal of Applied Econometrics*, 20(7), 873–889.

Kupiec, P. H. (1995). Techniques for verifying the accuracy of risk measurement models. *Journal of Derivatives*, 3(2), 73–84.

Moreira, A., & Muir, T. (2017). Volatility-managed portfolios. *Journal of Finance*, 72(4), 1611–1644.

Nystrup, P., Madsen, H., & Lindström, E. (2018). Dynamic portfolio optimization across hidden market regimes. *Quantitative Finance*, 18(1), 83–95.

Patton, A. J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1), 246–256.

Politis, D. N., & Romano, J. P. (1994). The stationary bootstrap. *Journal of the American Statistical Association*, 89(428), 1303–1313.

J.P. Morgan/Reuters (1996). *RiskMetrics—Technical Document*, 4th ed. New York.

Rydén, T., Teräsvirta, T., & Åsbrink, S. (1998). Stylized facts of daily return series and the hidden Markov model. *Journal of Applied Econometrics*, 13(3), 217–244.